Behavioral Economics in Sports

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Introduction

- **Behavioral Economics** is the study of how psychological, social, cognitive, and emotional factors affect economic decisions
- Sports offer a unique setting to study behavioral economics due to the high stakes, competitive nature, and large amounts of data
- We will explore the intersection between behavioral economics and sports through: Game Theory, Risk Attitudes, and Asset Valuation
- But first ... a brief history of behavioral economics!

History of Behavioral Economics I

- **The Invisible Hand and Laissez-Faire**
  Smith (1776) introduces the "invisible hand" and the fundamental idea of individuals' self-interest that then aggregate into markets

- **Homo Economicus**
  Mill (1836) introduces the "rational" decision-maker who maximizes wealth and simultaneously minimizes the associated costs

- **Rationality under Uncertainty**
  von Neumann & Morgenstern (1944) prove that preference relations satisfying certain axioms can be represented as utility functions such that agents' choices can be viewed as utility-maximizing

History of Behavioral Economics II

- **Bounds of Rationality**
  Simon (1955) highlights that individuals often only use limited information for their decisions that "satisfice" rather than "optimize"

- **Psychology and Economics**
  Kahneman & Tversky (1979) find major deviations from rationality under uncertainty due to agents placing more emphasis on losses

- **Modern Choice Architecture**
  Thaler & Sunstein (2008) introduce the concept of "nudging" to subtly influence decisions with minimal restrictions on choices

Outline

1. The Beautiful Dataset
2. Game Theory
3. Risk Attitudes
4. Asset Valuation

The Beautiful Dataset (Palacios-Huerta 2025)

- Recent manifesto written as a literature review / call-to-action that encourages studying economics using the context of sports
- "*Natural experiments, clean observability, precise measurement, high stakes, expert subjects, unimaginable detail, large datasets, exogenous rule changes, quasi-experimental variation, observable social effects, and no Hawthorne effects*"
- **Core:** Data Quality, Natural Experiments, and Identification

Palacios-Huerta (2025): Data Quality

- **Clean Observability**
  Matches are televised and recorded; highlights are available; statistical aggregators provide detailed data

- **Precise Measurement**
  Highly accurate measurements of player and team performance; advanced metrics available on platforms such as *SportsReference*

- **Large Datasets**
  Various sports played across dozens of countries; hundreds of games played each season for decades

Palacios-Huerta (2025): Natural Experiments

- **Natural Experiments**
  Laboratories are synthetic, do not readily align with reality; sports are a better representation of the real-world

- **Expert Subjects and High Stakes**
  Participants are professionals earning salaries in the millions; minimal concerns about subjects' objectives and requisite abilities

- **No Hawthorne Effects**
  Players **aware** of observation, but fully focused on the game; minimal concerns about changes in behavior due to observation

Palacios-Huerta (2025): Identification

- **Exogenous Rule Changes**
  Rule changes by governing bodies influence owners and players; Peeters, Szymanski, Fumagalli & Thomas (2014) examine Financial Fair Play in soccer

- **Quasi-Experimental Variation**
  Teams play in different settings against different opponents; Anderson, Rosen, Rust & Wong (2025) examine tennis serves

- **Observable Social Effects**
  Team sports feature many forms of strategic inter-person interactions; Anand & Rosen (2026) examine asymmetric influence of quarterbacks

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Introduction to Game Theory

- A **game** consists of **players** who perform **actions** to obtain **payoffs**
  - $\underbrace{\text{Ash and Bob}}_{\text{Players}}$ play $\underbrace{\text{rock-paper-scissors}}_{\text{Actions} = (\text{R, P, S})}$ to decide who $\underbrace{\text{pays for dinner}}_{\text{Payoffs}}$
- Games are subject to **rules** that determine the set of actions, payoffs, and strategies at each given state
  - R-P-S is a **simultaneous**, **static** and **complete information** game
- Crucially, payoffs for **each** player depend on actions of **all** players
  - Ash wins if she picks R, Bob picks S; loses if Bob picks P
- A **strategy** is a complete plan of actions for a player contingent on other players' actions
  - Ash plays R because she believes Bob will play S

Payoff Matrix for Rock-Paper-Scissors

		Bob		
		Rock	Paper	Scissors
	Rock	0	–1	+1
Ash	Paper	+1	0	–1
	Scissors	–1	+1	0

Table 1: Payoff Matrix for Rock-Paper-Scissors

- Each player picks their action simultaneously and independently
- Rock-paper-scissors is a **zero-sum** game where if a player wins, the other must lose; the combination of actions determines the payoffs (+1 for a win, –1 for a loss, 0 for a tie)

Nash Equilibria: Pure and Mixed

- A **Nash Equilibrium** is a set of strategies where no player can deviate from their own chosen action to improve their payoff
  - In a **pure** Nash Equilibrium, each player plays a specific action
  - In a **mixed** Nash Equilibrium, each player plays multiple actions with non-zero probabilities (a so-called "mixed strategy")
- Rock-Paper-Scissors has **no** pure Nash Equilibrium, only a mixed Nash Equilibrium where each player plays each action with probability $\frac{1}{3}$
  - Behavioral studies show that players often deviate from this equilibrium as rock is the most common choice by a wide margin (Costikyan 2013)

Examples of Game Theory in Sports

- **American Football**
  Ed Reed vs. Peyton Manning
- **Cricket**
  Steve Smith vs. Wahab Riaz
- **Soccer**
  Dean Henderson vs. Paddy McNair

Penalty Kicks in Soccer

- In soccer, a penalty kick is awarded when a defender commits a foul on an attacker within the penalty area (18 yds from the goal line)
- Kicker chooses a side to shoot; goalkeeper chooses a side to dive
- Ball travels at $\approx$ 70 mph (110 kmph) and reaction time $\approx$ 0.30s; Kicker and goalkeeper must anticipate each others' direction **before** the ball is kicked (actions are effectively simultaneous)
- Players prefer kicking in the direction opposite to their dominant foot (e.g. right-footed players prefer shooting left)

Penalty Kicks (Chiappori, Levitt & Groseclose 2002)

- Under **complete information** with the knowledge that kickers prefer their opposite side, the Nash Equilibrium implies certain probabilities of shooting and diving assuming the following hold:
  - Kickers prefer the side opposite to their footedness ("natural side"),
  - Kickers prefer the opposite of the goalkeeper's action (if known),
  - Scoring probabilities are higher on the natural side (generally L) (including in the case if the goalkeeper dives correctly)

		G		
		L	C	R
	L	P_L	π_L	π_L
K	C	μ	0	μ
	R	π_R	π_R	P_R

Table 2: Payoff Matrix in Penalties ($\pi > \mu > P$; $L > R$)

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,

Explanation

Ball travels too quickly to react; must anticipate opponent's action
No specific dominant direction for either player; must randomize

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,
- Scoring probabilities are roughly equal across each choice of direction,

Explanation

If the kicker has a "better" side, the kicker chooses this side more often, but the goalkeeper adjusts accordingly, which reduces the odds of scoring

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,
- Scoring probabilities are roughly equal across each choice of direction,
- Kickers choose L more often than choosing R,

Explanation

Kickers prefer their natural side as probability of scoring is higher

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,
- Scoring probabilities are roughly equal across each choice of direction,
- Kickers choose L more often than choosing R,
- Goalkeepers choose L more often than choosing R,

Explanation

Goalkeepers know kickers' shots will be towards kickers' natural side to maximize their odds of scoring, which encourages dives to the natural side

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,
- Scoring probabilities are roughly equal across each choice of direction,
- Kickers choose L more often than choosing R,
- Goalkeepers choose L more often than choosing R,
- Kickers choose L **less** often than goalkeepers,

Explanation

Kickers know goalkeepers anticipate shots towards the natural side, which encourages them to shoot towards the opposite side instead

CLG (2002): Individual Predictions

- Kickers and goalkeepers randomize independently,
- Scoring probabilities are roughly equal across each choice of direction,
- Kickers choose L more often than choosing R,
- Goalkeepers choose L more often than choosing R,
- Kickers choose L **less** often than goalkeepers,
- Kickers choose C more often than goalkeepers,

Explanation

Kickers already know to avoid C since goalkeepers can guarantee a save by choosing C, which means goalkeepers rarely need to choose C

CLG (2002): Individual Predictions

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- Kickers choose L more often than choosing R,
- Goalkeepers choose L more often than choosing R,
- Kickers choose L **less** often than goalkeepers,
- Kickers choose C more often than goalkeepers,
- Excluding all C options, (L, L) is the most common and (R, R) is the least common combination of player actions

Explanation

In general, both players are more likely to choose the left side due to the inherent natural side preference for kickers plus goalkeepers' anticipation

CLG (2002): Results in the Aggregate

- Previous kicks do **not** predict future kicks,
- Scoring probabilities roughly 75% for all kicks,
- Kickers chose L 206 times compared to R 174 times,
- Goalkeepers chose L 260 times compared to R 188 times,
- Kickers chose L 206 times compared to 260 times by goalkeepers,
- Kickers chose C 79 times compared to 11 times by goalkeepers,
- (L, L) occurred 117 times compared to 75 times for (R, R)

Observed behavior aligns with theoretical predictions!

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Risk Attitudes

- Risk attitudes describe how people respond to uncertainty
  - **Risk-Averse:** Prefer sure outcomes to risky ones
  - **Risk-Seeking:** Prefer risky outcomes to sure ones
  - **Risk-Neutral:** Indifferent provided expected utility is the same
- Expected utility theory can fail to model people's behavior
  - Allais, Ellsberg, and Rabin paradoxes
  - 4th downs in American football (Romer 2006)

Allais Paradox

- **Prizes:** ($0, $1M, $5M)
  - **Lottery 1A:** (0%, 100%, 0%)
  - **Lottery 1B:** (1%, 89%, 10%)
- **Question:** Do you prefer 1A or 1B?

Allais Paradox

- **Prizes:** ($0, $1M, $5M)
  - **Lottery 2A:** (89%, 11%, 0%)
  - **Lottery 2B:** (90%, 0%, 10%)
- **Question:** Do you prefer 2A or 2B?

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  - **Lottery 2A:** (89%, 11%, 0%)
  - **Lottery 2B:** (90%, 0%, 10%)
- Many people prefer 1A to 1B and 2B to 2A
  - 2A is 1A after subtracting 89% from probability of $1M
  - 2B is 1B after subtracting 89% from probability of $1M
- **Certainty Effect:** People like certainty of 1A

Ellsberg Paradox

- 300 balls in urn. 100 red, 200 yellow or blue
  - **Lottery 1A:** Win if red, lose if not
  - **Lottery 1B:** Win if blue, lose if not
- **Question:** Do you prefer 1A or 1B?

Ellsberg Paradox

- 300 balls in urn. 100 red, 200 yellow or blue
  - **Lottery 2A:** Win if yellow or red, lose if not
  - **Lottery 2B:** Win if yellow or blue, lose if not
- **Question:** Do you prefer 2A or 2B?

Ellsberg Paradox

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  - **Lottery 1A:** Win if red, lose if not
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  - **Lottery 2A:** Win if yellow or red, lose if not
  - **Lottery 2B:** Win if yellow or blue, lose if not
- Many people prefer 1A to 1B and 2B to 2A
  - If you prefer 1A to 1B, you think red more likely than blue
  - If you prefer 2B to 2A, you think blue more likely than red
- **Ambiguity Aversion:** People prefer known to unknown risks

Rabin Paradox

- **Question:** Would you play either of these lotteries?
  - **Lottery 1:** 50% you lose \$100; 50% you win \$110
  - **Lottery 2:** 50% you lose \$1,000; 50% you win \$1,000,000

Rabin Paradox

- **Question:** Would you play either of these lotteries?
  - **Lottery 1:** 50% you lose \$100; 50% you win \$110
  - **Lottery 2:** 50% you lose \$1,000; 50% you win \$1,000,000
- Many people do not play 1, almost everyone plays 2
  - Per expected utility theory, not playing 1 $\Rightarrow$ not playing 2
  - Holds even if \$1,000,000 becomes \$1,000,000,000 or anything else
- **Loss Aversion:** People hate losses more than they like gains

4th Downs in Football (Romer 2006)

- Do NFL coaches have rational risk attitudes on 4th down?
  - You get 4 chances ("downs") to advance the ball 10 yards
  - On 4th down, you can punt, kick a field goal, or "go for it"
- **Problem:** Must calculate value of field position
  - **Example:** 4th and goal from opponent's 4 yard line
  - If you go for it and fail, opponent gets ball at own 4

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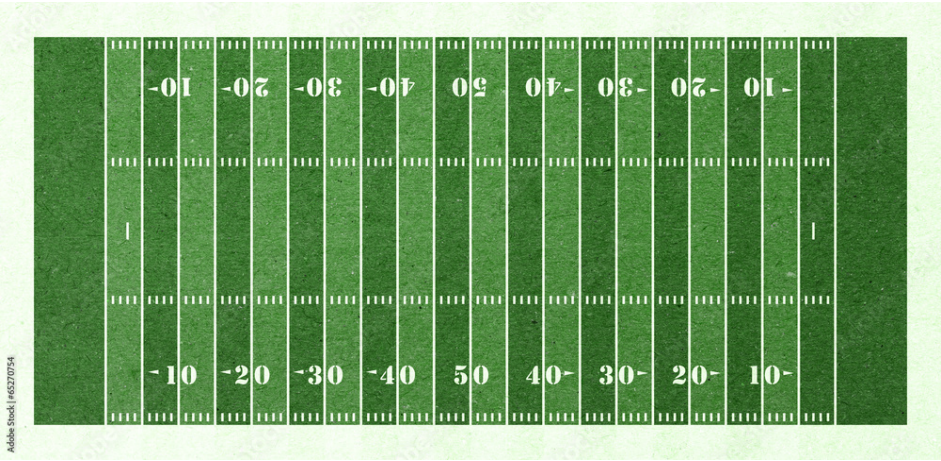

Figure 1: Football Field (dekART 2025)

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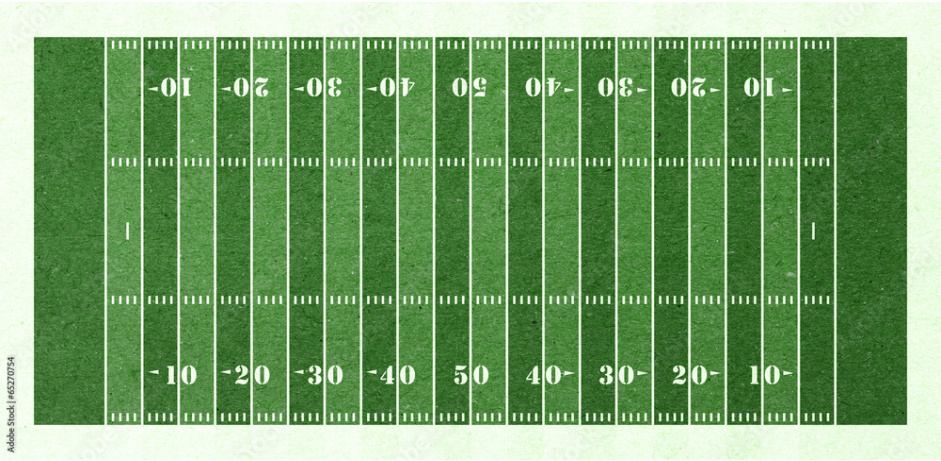

Figure 1: Football Field (dekART 2025)

- **Solution:** Dynamic programming (optimization)

Cake-Eating Problem

- You have 1 pound cake, how much to eat today vs. tomorrow?
- To solve recursively, suppose you'll have x cake left tomorrow
  - **Tomorrow:** Cake will go bad afterward, so eat all x of it
  - **Today:** Knowing this, can solve for how much to eat today
- Recursive approach essential when problem has many periods
  - Macroeconomists like Romer use dynamic programming to build models of long-run economic growth and business cycles
  - Due to a "midlife crisis" (Dash 2010), he applied it to football

Romer's Findings

- Coaches were too conservative on 4th downs
  - Of 1,604 4th downs where kicking was better, went for it 9 times
  - Of 1,068 4th downs where going for it was better, kicked 959 times
- Why were coaches hurting their win probability?
  - **Certainty Effect:** Punting certain to avoid turnover on downs
  - **Ambiguity Aversion:** May not know conversion probability
  - **Loss Aversion:** Hate failing more than like succeeding
- Coaches resisted change at first but have come around
  - Jim Fassel: "This is a professor from Cal-Berzerkely?" (Garber 2002)
  - Dan Campbell: "Wear a diaper before some of these games" (Patra 2023)
  - ESPN documentary on Romer's paper: "NFL and 4th Down"

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Asset Valuation

- Traditional assumptions: **rational expectations** and **efficient markets**
  - People use all available data to make unbiased forecasts of the future
  - Markets aggregate these forecasts into unbiased asset valuations
- In reality, markets do not always obey these assumptions
  - Winner's curse and psychological biases
  - Loser's curse in the NFL draft (Massey & Thaler 2013)

The Winner's Curse

- In common value auctions, the winner often overpays
  - Oil firms bidding on drilling rights (Capen, Clapp & Campbell 1971)
  - NFL teams trading up for a draft prospect (Massey & Thaler 2013)
- Why does this "winner's curse" happen?
  - The actual common value of the asset is unknown
  - Some firms are too optimistic, others too pessimistic
  - The most optimistic firm wins and therefore overpays
- **Solutions:** Acquire more information and bid more conservatively

Psychological Biases

- Winner's curse is "mechanical," but other biases are psychological
  - **Nonregressive Predictions:** People's predictions are more extreme and varied than the data suggest (Kahneman & Tversky 1973)
    - Scouts use the phrase "generational talent" too often
    - More 1st-rounders busts than Pro Bowlers (Massey & Thaler 2013)
  - **Overconfidence** (Alpert & Raiffa 1982)
    - More information $\Rightarrow$ greater overconfidence in judgments
    - Still, GMs must collect as much information as possible
  - **False Consensus** and **Anticipated Regret**
    - You like a player $\not\Rightarrow$ everyone else does (Ross, Greene & House 1977)
    - Fear of missing out on a potential superstar (Lerner & Tetlock 1999)

★ Look out for these biases when valuing assets!

Old Takes Exposed

Question: Which psychological bias does each quote represent?

1. Skip Bayless: "The Houston Texans, with this year's No. 1 overall pick, will forever regret it if they don't take the Texas kid with the movie-title nickname, Johnny Football." (Bayless 2014)
2. Dave Gettleman: "I know for a fact there were two teams that would've taken [Daniel Jones] in front of 17." (Benjamin 2019)
   - Ralph Vacchiano reported those teams were the Broncos and Redskins
   - Mike Klis countered that the Broncos would not have taken Jones
3. Mike Ditka: "We are going to win the Super Bowl. We got Ricky and he's the final piece to the puzzle." (Fleming 2019)

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Answer: (1) Anticipated Regret, (2) False Consensus, (3) Overconfidence

Loser's Curse in the NFL Draft (Massey & Thaler 2013)

- Does the market value of top draft picks match their performance?
  - **Reverse-Order Draft:** Worst teams pick first in each round
  - Teams can trade picks for current or future picks
- From 1983 to 2008, teams valued top picks highly
  - Trade value of last 1st-rounder was 50% of 10th, which was 50% of 1st
  - 1st pick JaMarcus Russell signed a 6 year, $61M contract in 2007
- Massey and future Nobel laureate Thaler were skeptical...

Linear Regression

- Suppose you want to predict something based on past data
  - If you study more, will your grades improve?
  - Are high-performing NFL players paid more?
- To perform linear regression, you:
  1. Define your dependent and independent variables
  2. Fit a line that best represents the correlation between the variables
  3. Use the line to predict future values of the dependent variable

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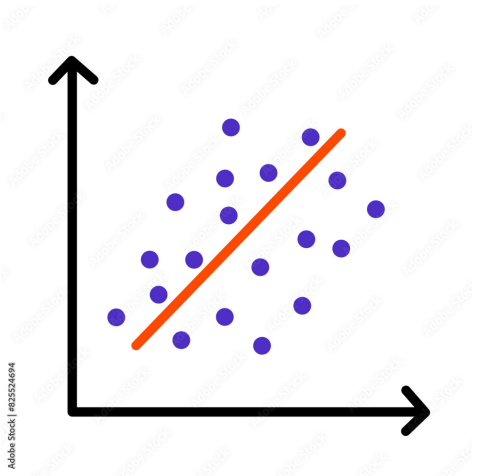

Figure 2: Linear Regression (Wira 2025)

Massey & Thaler's Results I

- **Analysis:** Trading up vs. down
  - Jimmy Johnson chart uses past trades to quantify each pick's worth
  - Calculate the number of starts and Pro Bowls each pick produced
  - Across all feasible 2-for-1 trades for 1st rounders, trading down yields 5.4 additional starts per season without a cost in Pro Bowls
- **Upshot:** Despite Massey & Thaler's results, teams still trade up
  - Ryan Pace: "If we want to be great, you just can't sit on your hands" (Wiederer 2017)
  - Howie Roseman: "The more chances you get, the more tickets to the lottery you get, the better you should be doing" (Berman 2015)
  - Brandon Beane: "If it doesn't work, the GM won't be here for the missing picks" (Walder 2024)

Massey & Thaler's Results II

- **Analysis:** Compensation for top picks
  - Regress Years 6–8 compensation on Years 1–5 performance
  - Years 1–5 Surplus = Predicted Compensation − Actual Compensation
  - Early 2nd round most valuable, and 1st pick least valuable of 1st round
- **Upshot:** Losing teams were "cursed" with top picks!
  - Massey: "The crime is thinking you can predict it" (Leonhardt 2020)
  - Since 2011, the NFL has greatly limited rookie compensation
    - 1st pick JaMarcus Russell signed a 6 year, $61M contract in 2007
    - 1st pick Cam Ward signed a 4 year, $48.7M contract in 2025
    - Salary cap was $109M in 2007 compared to $279.2M in 2025

Conclusion

- Sports are a great laboratory for behavioral economics
  - Cleaner objectives and data than real-world settings
  - Unlike lab subjects, pro athletes are experts and care about winning
- Game Theory, Risk Attitudes, and Asset Valuation
  - Pro soccer players play equilibrium strategies on penalty kicks
  - NFL coaches were too risk-averse on 4th downs
  - NFL general managers overvalued top draft picks

★ "It is not a question of what economics can do for sports, but what sports can do for economics" (Palacios-Huerta 2025)

References I

Allais, P. M. (1953). "Le Comportement de l'Homme Rationnel devant le Risque: Critique des Postulats et Axiomes de l'Ecole Americaine". In: *Econometrica* 21.4, pp. 503–546.

Alpert, M. & H. Raiffa (1982). "A progress report on the training of probability assessors". In: *Judgment under uncertainty: Heuristics and biases*. Ed. by D. Kahneman, P. Slovic & A. Tversky. Cambridge, UK: Cambridge University Press, pp. 294–305.

Anand, S. & J. Rosen (2026). "Resource Optimization Under Asymmetric Spillovers in the NFL". In: *Working Paper*.

Anderson, A., J. Rosen, J. Rust & K.-P. Wong (2025). "Disequilibrium Play in Tennis". In: *Journal of Political Economy* 133.1, pp. 190–251.

Bayless, S. (2014). "Texans have to take Manziel". In: *ESPN.com*.

Benjamin, C. (2019). "Dave Gettleman: 'I know for a fact' two teams would've picked Daniel Jones before No. 17". In: *CBS Sports*.

Berman, Z. (2015). "Howie Roseman — in 2015 — on trading up in the draft". In: *The Philadelphia Inquirer*.

Capen, E., R. Clapp & W. Campbell (1971). "Competitive Bidding in High-Risk Situations". In: *Journal of Petroleum Technology* 23.6, pp. 641–653.

Chiappori, P.-A., S. Levitt & T. Groseclose (2002). "Testing Mixed-Strategy Equilibria When Players Are Heterogeneous: The Case of Penalty Kicks in Soccer". In: *American Economic Review* 92.4, pp. 1138–1151.

Costikyan, G. (2013). *Uncertainty in Games*. Cambridge, MA: MIT Press.

Dash, E. (2010). "In the N.F.L. on Fourth Down, Romer Says Go for It". In: *The New York Times*.

References II

dekART (2025). *football field green paper*. Public Domain Image. URL: https://stock.adobe.com/images/football-field-green-paper/65270754.

Ellsberg, D. (1961). "Risk, Ambiguity, and the Savage Axioms". In: *The Quarterly Journal of Economics* 75.4, pp. 643–669.

Fleming, D. (2019). "Yep, the 1999 NFL draft was just as crazy as you thought". In: *ESPN.com*.

Garber, G. (2002). "Fourth-down analysis met with skepticism". In: *ESPN.com*.

Kahneman, D. & A. Tversky (1973). "On the psychology of prediction". In: *Psychological Review* 80.4, pp. 237–251.

Kahneman, D. & A. Tversky (1979). "Prospect Theory: An Analysis of Decision under Risk". In: *Econometrica* 47.2, pp. 263–291.

Leonhardt, D. (2020). "Hubris, a Case Study". In: *The New York Times*.

Lerner, J. S. & P. E. Tetlock (1999). "Accounting for the Effects of Accountability". In: *Psychological Bulletin* 125.2, pp. 255–275.

Massey, C. & R. H. Thaler (2013). "The Loser's Curse: Decision Making and Market Efficiency in the National Football League Draft". In: *Management Science* 59.7, pp. 1479–1495.

Mill, J. S. (1836). "On the Definition of Political Economy, and on the Method of Investigation Proper to It". In: *London and Westminster Review* 4.26, pp. 1–29.

Palacios-Huerta, I. (2025). "The Beautiful Dataset". In: *Journal of Economic Literature* 63.4, pp. 1363–1423.

Patra, K. (2023). "Lions HC Dan Campbell on fourth down calls: 'Just wear a diaper before some of these games'". In: *NFL.com*.

References III

Peeters, T., S. Szymanski, C. Fumagalli & C. Thomas (2014). "Financial fair play in European football". In: *Economic Policy* 29.78, pp. 343–390.

Rabin, M. (2000). "Risk Aversion and Expected-Utility Theory: A Calibration Theorem". In: *Econometrica* 68.5, pp. 1281–1292.

Romer, D. (2006). "Do Firms Maximize? Evidence from Professional Football". In: *Journal of Political Economy* 114.2, pp. 340–365.

Ross, L., D. Greene & P. House (1977). "The "false consensus effect": An egocentric bias in social perception and attribution processes". In: *Journal of Experimental Social Psychology* 13.3, pp. 279–301.

Simon, H. A. (1955). "A Behavioral Model of Rational Choice". In: *The Quarterly Journal of Economics* 69.1, pp. 99–118.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*. London: W. Strahan and T. Cadell.

Thaler, R. H. & C. R. Sunstein (2008). *Nudge: Improving Decisions About Health, Wealth, and Happiness*. New Haven, CT: Yale University Press.

von Neumann, J. & O. Morgenstern (1944). *Theory of Games and Economic Behavior*. Princeton, NJ: Princeton University Press.

Walder, S. (2024). "Seven clichés that show flaws in NFL draft trade-up process". In: *ESPN.com*.

Wiederer, D. (2017). "Everything Bears GM Ryan Pace said about trading up to draft QB Mitch Trubisky". In: *Chicago Tribune*.

Wira (2025). *Correlation icon vector illustration*. Public Domain Image. URL: https://stock.adobe.com/images/correlation-icon-vector-illustration/825524694?prev_url=detail.